Phraseology

CM 1 : Welcome and Introduction

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2 Introduction

3 Multiword Expressions

4 Multiword Expressions and NLP

About me



You will see me in...

- Prompt Engineering M2
- Software Projects M2
- Written Corpora M2

About this course

7 lectures

- Phraseology and Multiword expressions
- Defixation and Refixation
- Phraseology in NLP

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Evaluation mode

- Oral session (50 %)
- Final exam (50 %)

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Final exam : ???

References for this course

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What is Phraseology ?

Wikipedia Definition[1]

> Phraseology is the study of set or fixed expressions, such as idioms, phrasal verbs, and other types of multi-word lexical units (often collectively referred to as phrasemes), in which the component parts of the expression take on a meaning more specific than, or otherwise not predictable from, the sum of their meanings when used independently.

Phraseology studies fixed sequences of words.
Several words, but only one meaning.

[1] https://en.wikipedia.org/wiki/Phraseology

A bit of History

Phraseology is a young discipline

- First appearances: end of 19th - begining of 20th century
- Term coined by Charles Bally in 1909
- since 1970 : "phraseological boom" ()

Phraseology is at a crossroad between several disciplines

- Phraseology **is not** just a subdomain of syntax
- Phraseology **is not** just a subdomain of lexicology
- Phraseology **is** a central domain of language sciences ()

Terminological Nightmare

clause, conventional collocation, colligation, discourse structuring device, formulae idiom, irreversible binomial, phrase, prefab, quotation, simile, ...

Consequences

- Lots of confusions / repetitions
- Incompatible properties ()

For any given classification, there will always be objects that fall outside of it ().

Conventional names

- Idiom
- Phraseme
- Set phrase
- Fixed expression
- Multiword expression

During this class, we will mainly use the term "Multiword expression" (MWE).

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How to define a MWE ?

() definition

"phrases that function as units rather than as series"[2]

() key parameters of a MWE

- the nature of its components (either lexical entries and grammatical paterns)
- the number of components in it
- its frequency and its status (is it conventionalized ?)
- the distance between its components
- the lexical and syntactic flexibility of its components
- the semantic role of its components and its level of compositionnality

[2] "locutions qui ont le caractère d'unités plutôt que de séries"

Exercise Time !

Study the MWEs within the following sentences

- Max kicked the bucket
- Max and Marie kicked the bucket
- I will look these student numbers up
- I love them all, kith and kin alike
- My bad for adding fuel on the fire

Exercise Time !

Study the MWEs within the following sentences

- Max kicked the bucket
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Can you identify some properties shared by all these MWEs ?

MWE properties

No clear consensus, but some shared properties

1. A MWE has several components
2. A MWE is somewhat fixed
3. A MWE has a somewhat idiomatic meaning

Parameters: the things we use to study a MWE.
Properties: the rules supposedly shared among all MWEs.

1. A MWE has several components

A MWE contains several units

- In latin languages, 1 unit ≈ 1 word
- Word: a sequence of characters delimited by spaces

Limits

- *Hot Dog* (english): two words ?
- *Evlerimdekiler*[3] (turkish): one word ?
- *faire d'une pierre deux coups*[4] (french): several words ?

[3] those who are in the houses
[4] killing two birds with one stone

1. A MWE has several components

Max	kicked	the	bucket
Max	kicked the bucket		
Max	and	Marie	kicked the bucket

My	bad	for	adding	fuel	on	the	fire
My	bad	for	adding fuel on the fire				

2. A MWE is somewhat fixed

The components of a MWE are stables

- *Fish and Chips*
- *May the force be with you*
- *I will kill / killed two birds with one stone*

Limits

- Some MWEs can accept slight variations
- Some MWEs can accept word substitutions

2. A MWE is somewhat fixed

Variation	**MWE**	**valid ?**
Base	Max kicked the bucket	yes
Tense	Max is going to kick the bucket	yes
Adverb	Max will kick the bucket in a few hours	yes
Insertion	Max has unfortunatly kicked the bucket	yes
Passivation	The bucket of Max was kicked	no
Clivage	This is the bucket that Max kicked	no
Substitution	Max kicked the vessel / pot	no

3. A MWE has a somewhat idiomatic meaning

MWEs tend to be non-compositional

- Several units, one meaning
- MWE Meaning $\neq$ sum of its components

Limits

- Semi-compositional MWEs (*adding fuel on the fire*)
- Fully-compositional MWEs (*to make a decision*)

3. A MWE has a somewhat idiomatic meaning

Semantic opacity of a MWE

- You can not guess a MWE meaning from its components
  - your MWE is non-compositional
- You can partially guess a MWE meaning from some of its components
  - your MWE is semi-compositional
- You can guess a MWE meaning from its components
  - yout MWE is fully-compositional

Reality: it's a *continuum*.

Last property: the most important one.

Take your guesses !

4. A MWE is Conventionalized

Conventionality

- Conventionalized MWE: known by a group of individuals
- **Conventionality is a trait shared by all MWEs**

About the notion of "group of individuals"

- The population of a region, a state, a country, etc
- The individuals inside this class

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Let's start with a question.

Do you think it is useful to process MWEs in NLP ?

MWEs in NLP

Interests

- Better contextualization
  - (): XX % of MWEs in english texts
  - Can resolve many ambiguities
- Human writing VS machine writing
  - Does LLMs use fewer MWEs than humans ?
- (Much) better translations
  - hang on ! ⇒ pends-toi
  - it's raining cats and dogs ⇒ il pleut des chats et des chiens

Two MWE-related tasks in NLP: discovery and identification.

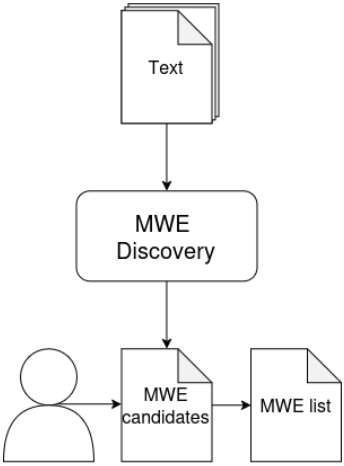
MWE Discovery

Goals

- Discover MWEs from texts
- No prior knowledge on MWEs

Methods

- N-gram frequency analysis
- Manual annotation of all candidates



MWE Discovery with Association Measure ()

$$I(x,y) = log^2 \frac{P(x,y)}{P(x)P(y)}$$

computer	scientist	1.12
keeping	from	2.14
set	up	7.30

a	doctor	0.96
set	about	-0.60
man	women	-0.12

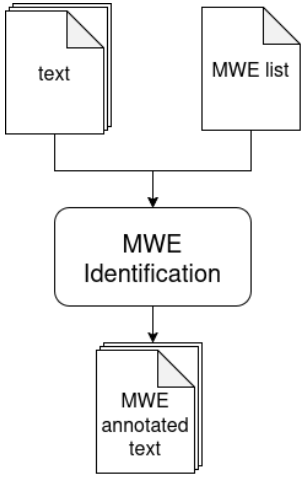
MWE Identification

Goals

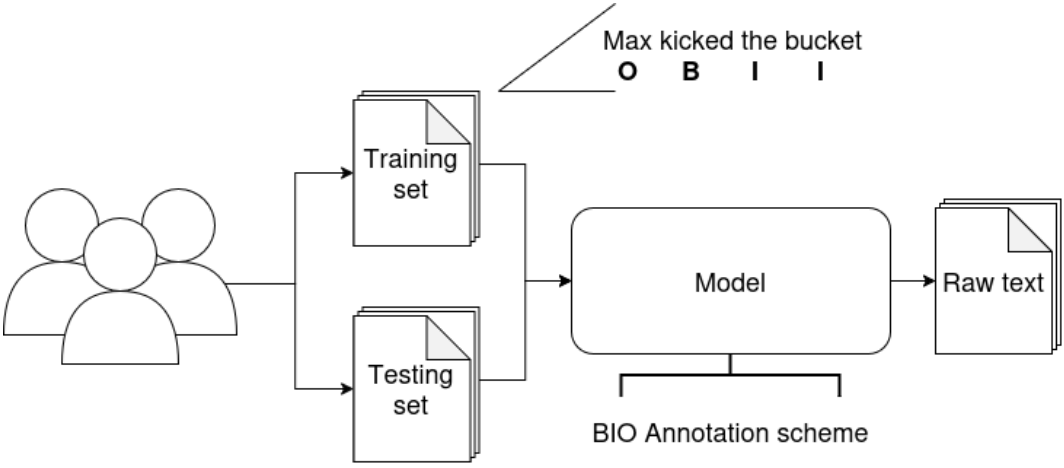
- Annotate MWEs in texts
- Take variation into account

Methods

- Rule-based or statistical
- Based on a list of expected MWEs



MWE Identification with Statistical Approaches



Exercise Time !

What about rule-based MWE identification ?

Find some rules which might help identificating MWEs in texts.

MWE Identification with Rule-based Approaches

Seen2Seen set of rules ()

- Use lemmas of each word
- Use POS tags of each word
- Consider discontinuities (maximum distance between units ?)
- Always favor closest components
- Units must have syntactic connections
- Consider lexical variations
- Consider nested MWEs (within each others)

Doggy bag

Multiword Expressions

- Studied in phraseology
- Several components
- Somewhat fixed
- Somewhat idiomatic
- Always conventionalized

MWEs and NLP

- MWE discovery
- MWE identification

